

Access Logs and Activity Summary

Suspect 1

Ayan Farooqi (Junior AI researcher)

- 2 failed attempts
- At 1:22- 1:24, he was flagged during login
- Might be the unknown user

Breach Clues Tracker

Category	What to Capture	Notes
System Components	Key parts of the system handling the data	The system components include: /raw/genomes , which is the folder with the genomes, the scheduled bot that shifts the files "SystemBot01" , and the genetic data uploaded from the clinic
Who has access	Who can see or use sensitive data(Read and write access)	The employees of Genixa have access to the information, including: (Cybersecurity lead) , Alex Mehta, Ayan Farooqi (Junior AI Researcher) , Riya Kapoor (Engineer) , Rina Solis (Head of Engineering) , and lastly, Sameer Joshi (Platform Architect)
Access Issues	Any red flags in how access is managed	Access is given to a junior researcher (Ayan Farooqi) , and he tried to login during odd hours and was "flagged off" ; hence, the main suspect blame goes to Ayaan Farooqi
Data Flow	How data moves through the system	Partner clinics upload genetic data through a secure portal to GenixaCloud , and these are stored in a /raw/genomics . Then every 6 hours, the uploaded files are taken from the /raw/genomics folder to the AI model input folder. There, the AI model reads the inputs and generates a risk score for the individual whose data it was given. The score is sent to the clinics.
Weak points	Parts that seem vulnerable or risky	The main part that is vulnerable and very important to remain confidential is (/raw/genomes) and (/model/input) because this is the place where all the genetic information is stored, and many employees have access to this confidential information.
Suspicious Activity	Anything unusual or flagged in the system	In the access logs, on Friday [10:11 AM] , Riya tried to access the .../raw/genomes folder, but in the file Genixa_System_Overview_Document at the same time, Ayan said "denied access message for a genome folder I normally use." Additionally, there are no logs from Friday, as it jumps to Wednesday. Furthermore, a login attempt that was shown during Thursday [02:02 PM] , which showed a login attempt Unknown_User accessing unknown_device from an unknown location .
Stakeholder	Key worries	One of the major concerns is that tampering with the

Concerns	raised	model could reduce the AI’s efficiency, but even more pressing for stakeholders is the risk of data leaks, which could erode client trust and lead to a loss of customers.
Early Suspicions	Where do you think the breach may have started?	The breach may have started early morning when “ Ayan Farooqi ” (Junior AI researcher) logged in to read the raw genome files at a very suspicious time, and then went on to write in the model input, which is only supposed to be edited by the scheduled bot “ SystemBot01. ”

Make a Short Observation Note Summarising:

There are quite a few unusual things that are in the logs, as listed below:

1. Usage of VPN: **Ayan Farooqi** used a **VPN** instead of using an Office IP.
2. Using Wrong Model: The model that he changes matches the file (**/model/input**) involved in the alert
3. Mismatch in Timeline: Access Denied and Access Granted are shown for **Riya Kapoor's** user on her PC, as mentioned in the file [**Access_Logs___Activity_Summary**], but the messages were from **Ayan Farooqi**, as shown in the [**Genixa_System_Overview_Document**].
4. Login Attempts: An **Unknown_User** from an **unknown_device** attempted to log in; however, they failed.
5. Log Date Mismatch: In the logs, the days mentioned are "**Thursday**", "**Friday**", and goes directly to "**Wednesday**"; there is no mention of "**Saturday**", "**Sunday**", even though they are not considered working days, the logs for "**Monday**" and "**Tuesday**" are also not mentioned; hence, that is a crucial anomaly and huge gap.

We also consulted **Ayan's** story, and initially, **Ayan's** story matches the timeline; however, as he continues his story, his story deviates from the logs. He tries to log in not to access the information, but just because he received an email; hence, to check if everything is fine and in place. Hence, that is why in the **Breach Clues Tracker Table**, we tried to prove that **Ayaan Farooqi** is one of the **biggest suspects** in the entire case.

Riya Kapoor's first access attempt failed, but then **Ayan** messaged **IT support**, and **Riya Kapoor** was granted access. We suspect that Ayan was secretly using Riya's account to access files. The question arises, 'Why was **Ayan** trying to log in from **Riya's** user?'

Task 3

Close the loop

Breach Timeline:

- **Ayan Farooqi** tried to access the **confidential information** at **1:22 AM**. He read the **genome files** and then wrote in the model input.
- He was **flagged** off for that. To prevent being **flagged** further, he again logged in at **10:12 am**, but he described that he was logging in, but the access log showed **Riya Kapoor**.
- This explains that he tried to log in from a different PC, so that his system does not reflect again, and is identified as the culprit.
- He reset the **GAccess** and managed to log in somehow at **10:12 AM**
- Only later, at **10:17 AM**, did he actually inform the **IT support** that he wasn't able to log in.

Immediate Fixes:

- **Give Appropriate Roles:** The suspect, **Ayan Farooqi**, was given access to **Read/Write** as a **junior researcher**. He should be given access to a secluded part of it under a **senior manager** instead. Moreover, he had no role in the **genetic information**, so why did he have to check the **genetic information**?
- **Centralised Access Oversight Implemented:** All access to sensitive data should be routed through a designated senior authority, and immediate detection of irregular activity.
- **Fixes Log Files:** The log files should be updated regularly to ensure timelines

Genexia's Next Step

- First of all, **Ayan Farooqi** has to be given a **termination notice** for whatever actions conducted by him.
- The **cops** should be informed immediately, and **Ayan Farooqui** should be handed to the **cops** with immediate effect.
- **The main lead team** has to go through the access logs and remove all kinds of access to the people who do not play any role in the **confidential** aspect of the **role**
- Employees with a **junior role** should be exempted from the **important confidential information**, or a **head** should be appointed to make sure any breach does not take place in the future.
- **Personal systems** should not be allowed in the **company's materials**; all important work should take place under **Office networks** and **office-based devices**

If-based rule:

```
logs = [  
  {"time": "01:13 AM", "user": "SystemBot01", "device_id": "auto_runner",  
   "location": "internal", "action": "Write", "folder": "/model/input"},  
  {"time": "01:22 AM", "user": "Ayan Farooqi", "device_id": "dev_ayan_lap",  
   "location": "Pune (VPN)", "action": "Read", "folder": "/raw/genomes"},  
  {"time": "01:24 AM", "user": "Ayan Farooqi", "device_id": "dev_ayan_lap",  
   "location": "Pune (VPN)", "action": "Write", "folder": "/model/input"},
```

```
{ "time": "04:42 PM", "user": "Ayan Farooqi", "device_id": "dev_ayan_lap",  
"location": "Office IP", "action": "Restart Model", "folder": "-"},  
{ "time": "10:11 AM", "user": "Riya Kapoor", "device_id": "eng_rkapoor_pc",  
"location": "Office IP", "action": "Access Denied", "folder": "/raw/genomes"},  
{ "time": "02:02 AM", "user": "Unknown_User", "device_id": "unknown_device",  
"location": "Unknown", "action": "Login Attempt", "folder": "-"}  
]
```

```
# Scan logs and flag anything suspicious
```

```
for entry in logs:
```

```
    time = entry["time"]
```

```
    user = entry["user"]
```

```
    action = entry["action"]
```

```
    folder = entry["folder"]
```

```
    location = entry["location"]
```

```
    device = entry["device_id"]
```

```
# 1. Flag actions that happen late at night
```

```
if "AM" in time and time < "05:00 AM":
```

```
    print(f"⚠ Suspicious time: {user} performed '{action}' at {time}")
```

```
# 2. Flag any unknown user
```

```
if user == "Unknown_User":
```

```
    print(f"⚠ Unknown user tried to log in at {time}")
```

```
# 3. Flag if access was denied
```

```
if action == "Access Denied":
```

```
    print(f"⚠ {user} was denied access to {folder} at {time}")
```

```
#---- NEW CODE ---
```

```
# 4. If VPN is used,
```

```
if "VPN" in location:
```

```
    print(f"⚠ {user} is accessing using a VPN in {location}")
```

```
# 5. Check if a personal device is used
```

```
if "lap" in device: # He tried to access from his personal laptop
```

```
    print(f"⚠ {user} is trying to access from personal device {device} at  
{location}")
```